Check in

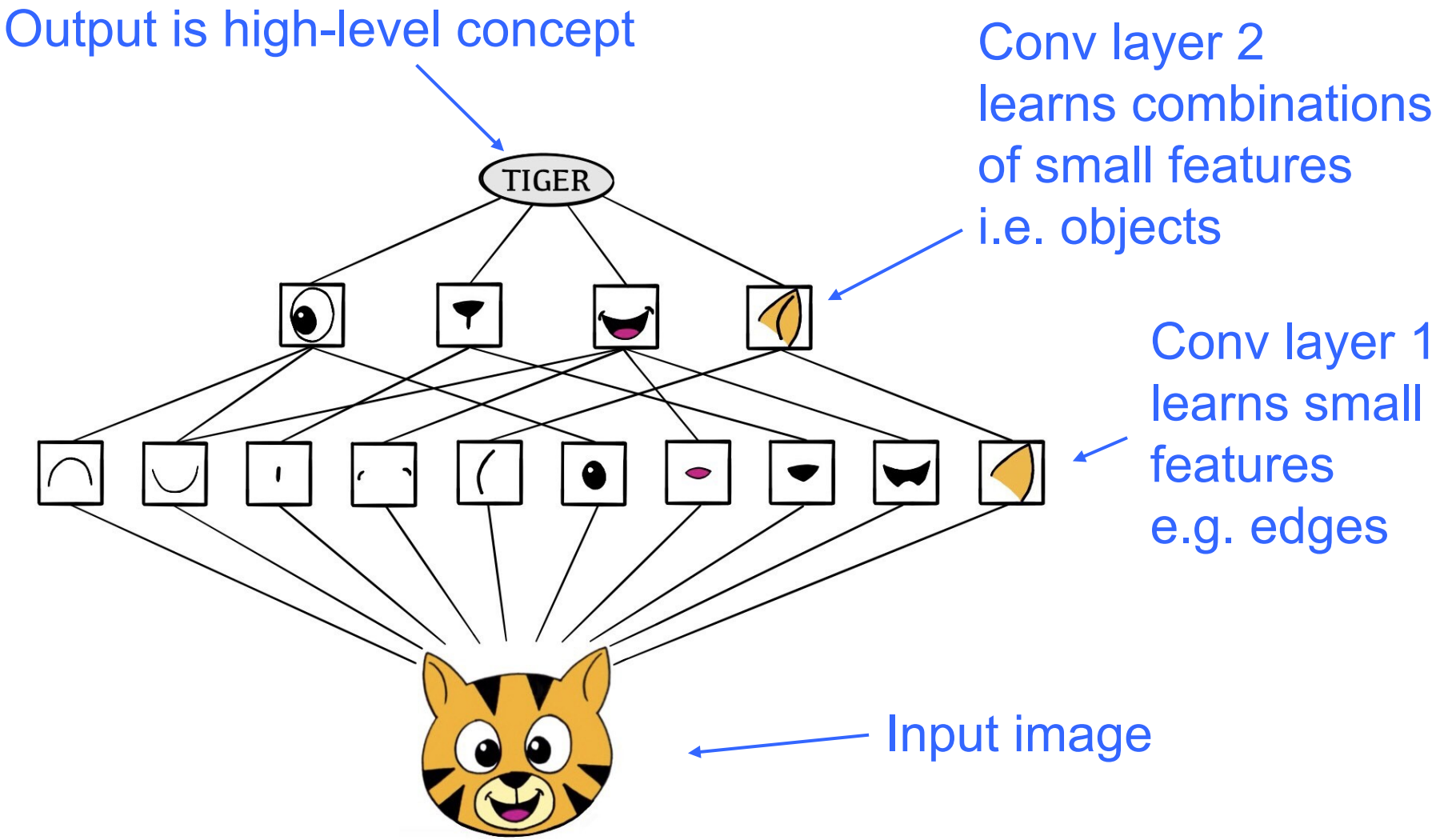
- How’s the assignment going?
- Coming up in second half of semester
  - Reading
  - Individual projects

Today

- Neural networks and deep learning
  - Single layer neural networks
    - using keras package
    - architectures for different questions
      - regression, classification, multifunction
  - Multi-layer neural networks
  - Convolutional neural networks
  - U-net
  - Transformers

deep

Convolutional NNs



ISLR 10.6

Convolutional NNs

- Example: CIFAR100
  - standard benchmark dataset: 100 categories
  - ecology subset: 56 categories
  - “ecoCIFAR56”
- Code: convolutional_nnet.R

Convolution layer

Input image matrix

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \\ j & k & l \end{bmatrix}$$

Convolution filter

$$\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$$

convolve

Output matrix

$$\begin{bmatrix} a\alpha + b\beta + d\gamma + e\delta & b\alpha + c\beta + e\gamma + f\delta \\ d\alpha + e\beta + g\gamma + h\delta & e\alpha + f\beta + h\gamma + i\delta \\ g\alpha + h\beta + j\gamma + k\delta & h\alpha + i\beta + k\gamma + l\delta \end{bmatrix}$$

ISLR p 413

Convolution layer

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ISLR p 413

Convolution layer

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Convolution filter

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weights to be trained

convolve

Dimension determined by rows & cols of the filter windows

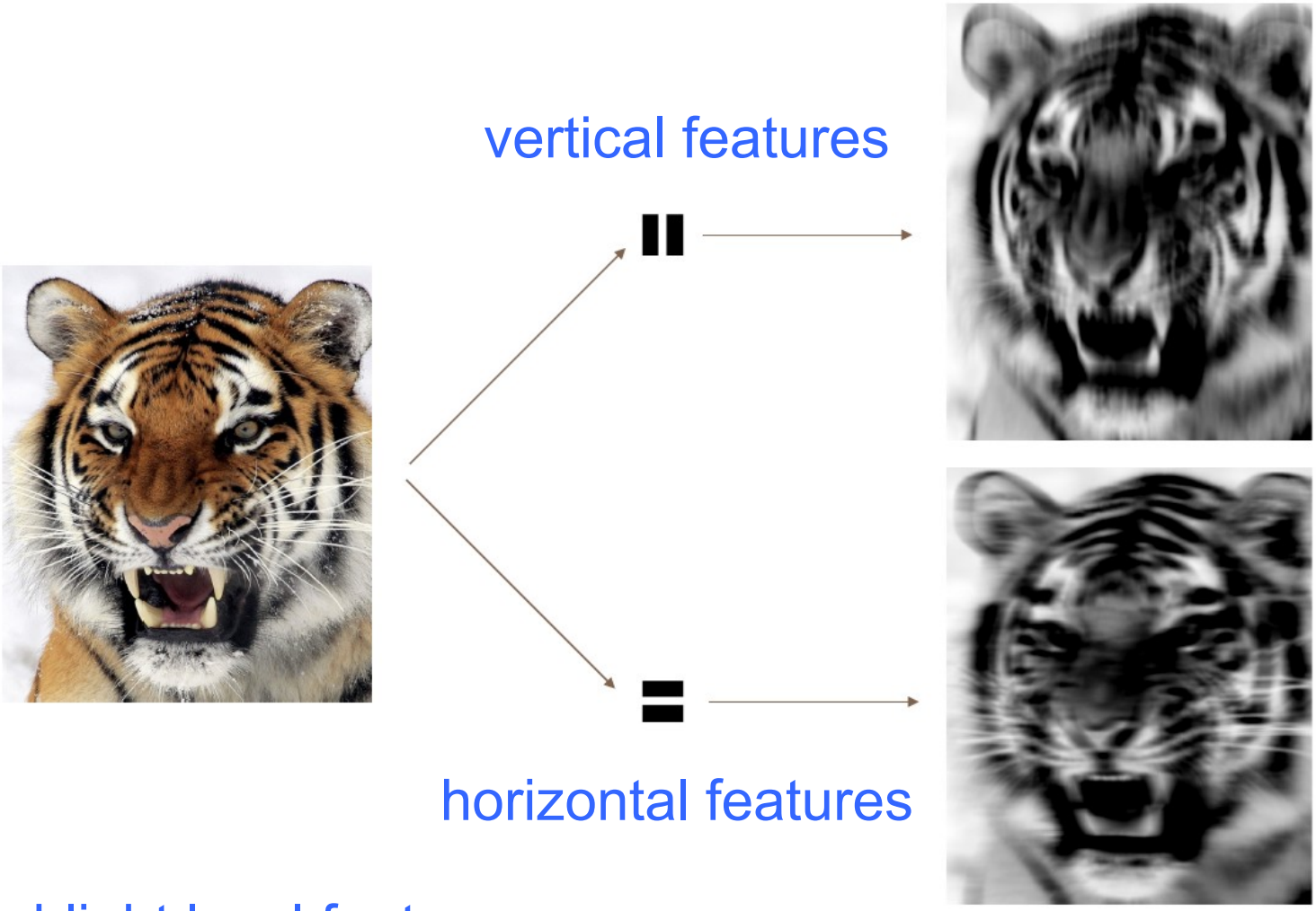
Output matrix

$$\begin{bmatrix} a\alpha + b\beta + d\gamma + e\delta & b\alpha + c\beta + e\gamma + f\delta \\ d\alpha + e\beta + g\gamma + h\delta & e\alpha + f\beta + h\gamma + i\delta \\ g\alpha + h\beta + j\gamma + k\delta & h\alpha + i\beta + k\gamma + l\delta \end{bmatrix}$$

ISLR p 413

Convolution layer

input image | filter | convolved image



filters highlight local features

ISLR 10.7

Pooling layer

Condenses information

e.g. “Max pooling”

$$\begin{bmatrix} 1 & 2 & 5 & 3 \\ 3 & 0 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 1 & 2 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$$

Input matrix
4x4

Output matrix
2x2

ISLR p 415

Pooling layer

Condenses information

e.g. “Max pooling”

Max

$$\begin{bmatrix} 1 & 2 & 5 & 3 \\ 3 & 0 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 1 & 2 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$$

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Output matrix
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ISLR p 415

Pooling layer

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e.g. “Max pooling”

Max

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ISLR p 415

Pooling layer

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ISLR p 415

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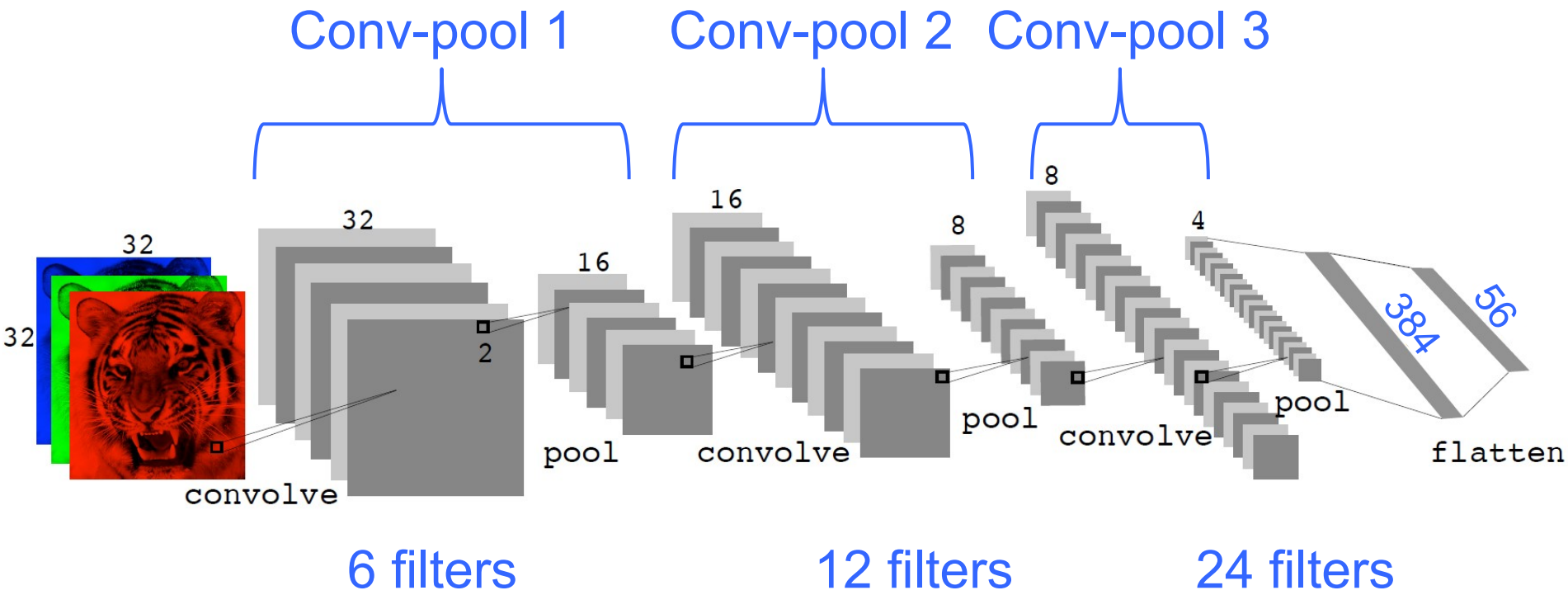
Max

Input matrix
4x4

Output matrix
2x2

ISLR p 415

Architecture



ISLR 10.8

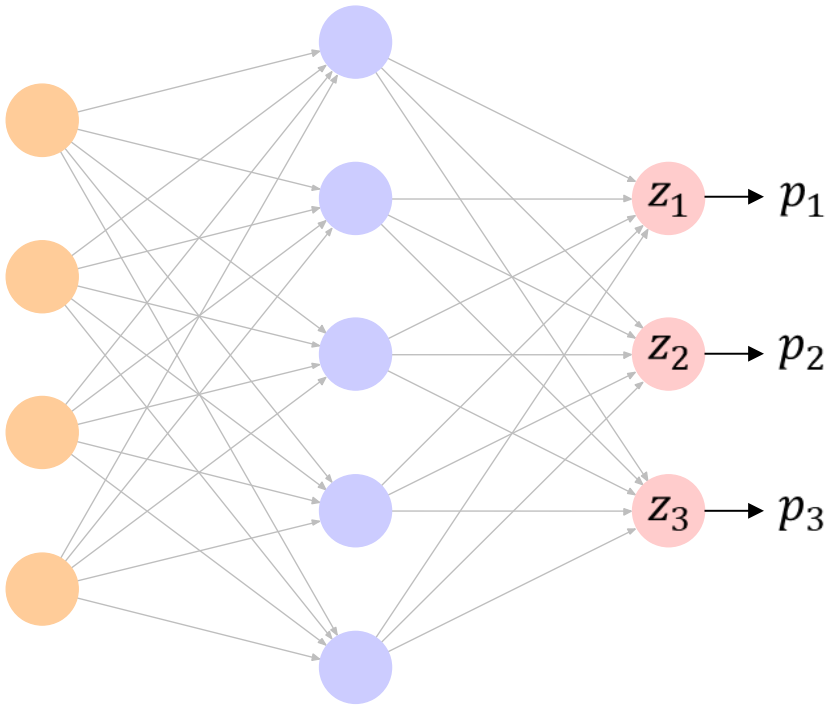
Softmax

Classification with multiple classes

Input k numbers (z_i) → output k probabilities (p_i)

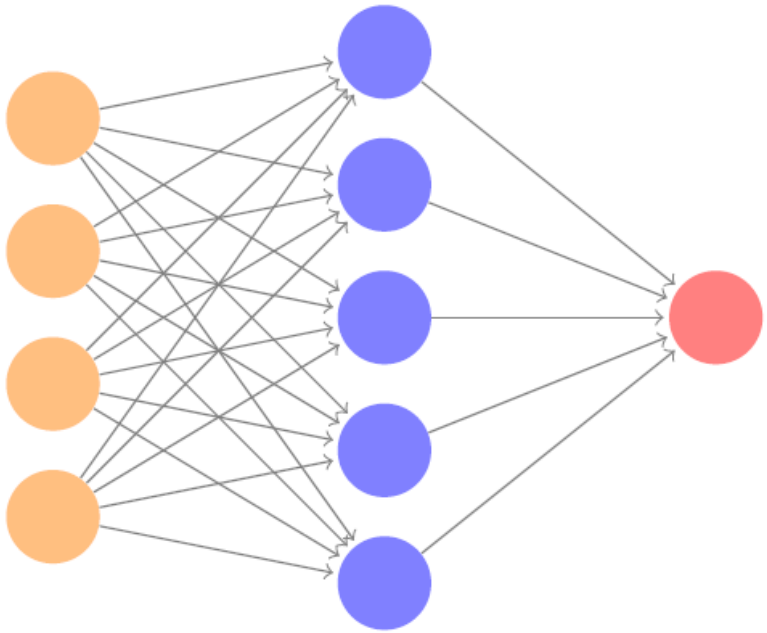
Probabilities sum to 1

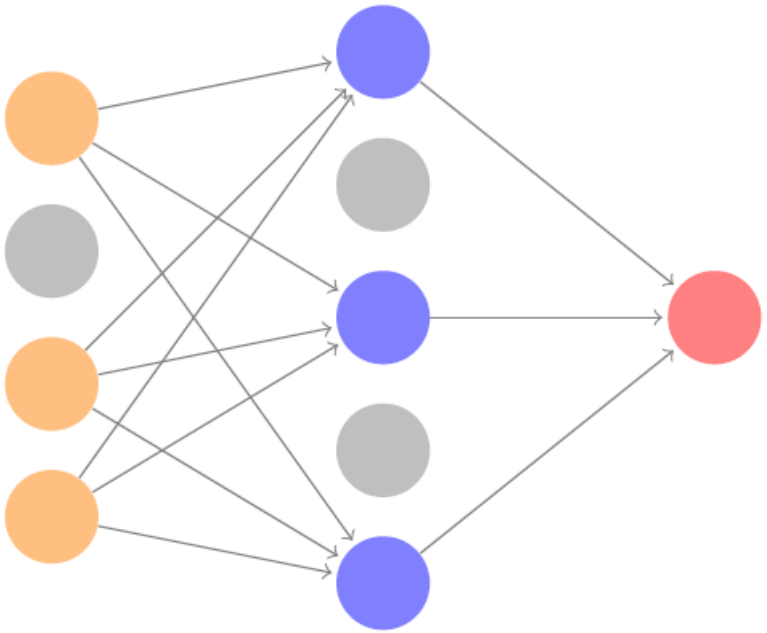
Exponential emphasizes the larger inputs



$$p_i = \frac{e^{z_i}}{\sum_{j=1}^{k} e^{z_j}}$$

Training algorithm: dropout





Like random forest sample columns

ISLR 10.19